

HADROS3: Physics and Statistical Theory of the Implemented Pipeline

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Implemented theory through H3-W10 PYTHIA 8 and HepMC3 Event Generation

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This document describes only the physics, statistics, numerical methods, and proxy models implemented in the current HADROS3 pipeline. It is not a user manual, dashboard guide, installation guide, or proposal for future stages. Where the implementation uses diagnostic renderers or physics proxies, that status is stated explicitly. This is a living theory document for HADROS3. It describes exclusively the physics currently implemented in the official HADROS3 pipeline. Future or planned models are not described here until they become part of the official implementation.

Official Physics Reference

This document is the official reference describing the physics currently implemented in HADROS3.

If any discrepancy is found between the implementation and this document, either:

- the implementation must be corrected; or
- this document must be updated.

Such discrepancies must never be ignored.

The scientific traceability chain for implemented physics is

$$\text{code} \rightarrow \text{provenance} \rightarrow \text{Theory PDF}. \quad (0.1)$$

Each simulation provenance record must identify the theory version that describes its physical and statistical assumptions.

Theory Version Information

Field	Value
Software Version	0.9.0
Physics Version	1.3
Theory Version	1.3
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Current implementation status	implemented through H3-W10 PYTHIA 8 and HepMC3 Event Generation

Stage	Theory status
H3-W5 UHE Source	implemented and documented
H3-W6 Forward Geodesics	implemented and documented
H3-W7 DIS Interaction Sampler	implemented and documented
H3-W8 Observer Bridge	implemented/proxy and documented
H3-W8b Observer Image Branch	implemented/proxy and documented
Analysis	
H3-W9a POWHEG Dry Run	implemented/dry run and documented
H3-W9b POWHEG Real Smoke/Free	implemented/smoke+free and documented
H3-W10 PYTHIA 8 + HepMC3 Event Generation	implemented and documented
H3-W11 GEANT4	implemented for supported-energy validation; UHE guarded
H3-W12 Photon Transport	planned, not documented as implemented
H3-W13 Spectra	planned, not documented as implemented

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Scope and Notation

The implemented HADROS3 pipeline is organized as a staged calculation,

$$\text{inputs} \rightarrow \text{run}() \rightarrow \text{outputs} \rightarrow \text{diagnostics} \rightarrow \text{provenance}. \quad (0.2)$$

Each implemented stage consumes official products from earlier stages and writes to its own output directory. The stages covered here are the camera preview, Kerr geometry utilities, analytic medium, UHE source, forward null geodesics, DIS optical-depth sampler, Observer Bridge scoring, and POWHEG dry-run/real hard-process generation, and PYTHIA 8 event generation with HepMC3 output.

This document is maintained as a required living reference for implemented physics. When a new HADROS3 stage changes or adds physical equations, statistical weights, physical approximations, or proxy models, this source file and the regenerated PDF must be updated with the implementation.

Table 3: Implemented-stage coverage in this theory document. Planned stages are listed only to mark that their physics is not yet documented as implemented.

HADROS3 stage	Implementation status	Theory documented
H3-W5 UHE Source	implemented	yes
H3-W6 Forward Geodesics	implemented	yes
H3-W7 DIS Interaction Sampler	implemented	yes
H3-W8 Observer Bridge	implemented/proxy	yes
H3-W8b Observer Image Branch Analysis	implemented/proxy	yes
H3-W9a POWHEG Dry Run	implemented/dry run	yes
H3-W9b POWHEG Real Smoke/Free	implemented/smoke+free	yes
H3-W10 PYTHIA 8 + HepMC3 Event Generation	implemented	yes
H3-W11 GEANT4	implemented/supported-energy only	yes
H3-W12 Photon Transport	planned	no
H3-W13 Spectra	planned	no

All radii in the dynamical Kerr calculation are expressed in gravitational radius units,

$$r_g = \frac{GM}{c^2}. \quad (0.3)$$

The implementation uses Boyer-Lindquist coordinates $x^\mu = (t, r, \theta, \phi)$ and the metric signature $(-, +, +, +)$. The dimensionless Kerr spin is denoted by a in units of r_g . The code clamps spin-dependent operations away from the extremal singular limit where needed.

Table 4: Primary symbols used in the implemented theory.

Symbol	Meaning
r, θ, ϕ	Boyer-Lindquist spherical coordinates in r_g units.
Σ, Δ, A	Standard Kerr metric functions.
p_μ	Covariant four-momentum components.
$E_{\nu, \text{local}}$	Neutrino energy measured in a local medium or ZAMO frame.
$\rho(r, \theta)$	Analytic torus rest-mass density.
n_b	Baryon number density, $n_b = \rho/m_b$.
$\sigma_{\nu N}(E)$	Tabulated charged-current neutrino-nucleon DIS cross section.
τ	Integrated DIS optical depth along one forward path.
S_{obs}	Observer Bridge final observation score.
x	Bjorken momentum fraction of the struck parton.
y	DIS inelasticity, $y = P \cdot q / (P \cdot k)$.
Q^2	Virtuality of the exchanged boson, $Q^2 = -q^2 > 0$.
W	Invariant mass of the produced hadronic system.
$\sqrt{s}$	Center-of-mass energy of the neutrino-nucleon system.
G_F	Fermi constant.
m_W	W boson mass.
$V_{q_i q_j}$	CKM matrix element for the quark flavor transition $q_i \rightarrow q_j$.
F_2, xF_3, F_L	CC neutrino DIS structure functions.
$Y_\pm$	DIS angular factors, $Y_\pm = 1 \pm (1 - y)^2$.
$B(\Phi_n)$	Born differential cross section on n -body phase space.
$\bar{B}(\Phi_n)$	NLO-corrected Born (virtual + integrated real counterterms).
$\Delta(\Phi_n, p_T)$	POWHEG Sudakov form factor.

Chapter 1

Camera / Kerr Ray Tracing

1.1 Implemented camera modes

The camera preview layer is a visual and diagnostic stage. It is separate from the DIS interaction physics and from the Observer Bridge scoring. Its main product is a rendered observer-view image, typically `CameraPreview/hadros3_camera_preview.png`. The implemented backend selection includes:

- **analytic_geometry_only**: always available; a schematic camera preview, not a physical Kerr ray-traced image.
- **kerr_like_cuda**: a self-contained HADROS3 CUDA preview executable using a Kerr-like camera model.
- **full_kerr**: a self-contained HADROS3 CUDA preview executable using the full-Kerr preview mode when available.
- **legacy_cpu_geodesic_preview**: detected for provenance but not used as the primary self-contained HADROS3 runtime path.

The camera preview can render a torus-like visual object and polar funnels. This visual object is not the DIS optical-depth calculation. The provenance marks camera-preview products separately from the shared `MediumRenderer`. In current provenance, the camera preview records `medium_renderer_used = false` because its purpose is visual camera preview, not the shared diagnostic density-map renderer.

1.2 Camera frame

Observer Bridge camera diagnostics use the same geometric convention for the camera direction that is also relevant when overlaying candidates on the camera preview. A camera position is built from

$$\mathbf{x}_{\text{obs}} = r_{\text{obs}} \begin{pmatrix} \sin i \cos \varphi_{\text{obs}} \\ \sin i \sin \varphi_{\text{obs}} \\ \cos i \end{pmatrix}, \quad (1.1)$$

where r_{obs} is `observer_distance_rg`, i is `observer_inclination_deg`, and φ_{obs} is `observer_azimuth_deg`. The forward axis points approximately toward the origin,

$$\hat{\mathbf{f}} = -\frac{\mathbf{x}_{\text{obs}}}{|\mathbf{x}_{\text{obs}}|}. \quad (1.2)$$

The right and up axes are built from a world z axis, with a fallback right axis if the camera is too close to the polar direction:

$$\hat{\mathbf{r}}_{\text{cam}} = \frac{\hat{\mathbf{f}} \times \hat{\mathbf{z}}}{|\hat{\mathbf{f}} \times \hat{\mathbf{z}}|}, \quad (1.3)$$

$$\hat{\mathbf{u}}_{\text{cam}} = \frac{\hat{\mathbf{r}}_{\text{cam}} \times \hat{\mathbf{f}}}{|\hat{\mathbf{r}}_{\text{cam}} \times \hat{\mathbf{f}}|}. \quad (1.4)$$

1.3 FOV diagnostic and Kerr-matched overlay

The Observer Bridge geometric camera view and FOV score use a pinhole proxy. It converts a candidate position $\mathbf{x}$ into a camera vector

$$\mathbf{d} = \mathbf{x} - \mathbf{x}_{\text{obs}} \quad (1.5)$$

and projects it as

$$z_{\text{cam}} = \mathbf{d} \cdot \hat{\mathbf{f}}, \quad (1.6)$$

$$x_{\text{ndc}} = \frac{\mathbf{d} \cdot \hat{\mathbf{r}}_{\text{cam}}}{z_{\text{cam}} \tan(\text{FOV}_x/2)}, \quad (1.7)$$

$$y_{\text{ndc}} = \frac{\mathbf{d} \cdot \hat{\mathbf{u}}_{\text{cam}}}{z_{\text{cam}} \tan(\text{FOV}_y/2)}. \quad (1.8)$$

The field-of-view flag is true when $z_{\text{cam}} > 0$ and $|x_{\text{ndc}}| \leq 1$, $|y_{\text{ndc}}| \leq 1$. This proxy is not the default mapping used by the camera overlay.

The default overlay records `candidate_overlay_projection_model=kerr_geodesic_pixel_match`. It integrates backward Kerr camera rays on the configured pixel grid and associates a candidate with the ray of acceptable minimum spatial approach. Thus the plotted pixel includes Kerr lensing of the camera ray. This remains a diagnostic position match, not secondary-particle emission or transport. The explicit `candidate_overlay_mapping=geometric_proxy` option retains the pinhole projection only as a labeled fallback.

Camera Preview stores raw PNG rows with $+e_\theta$ at the top, which is south-up for the adopted spherical basis. Every spatial Observer Overview product applies exactly one vertical display transform, `flip_y_for_north_up`; its background and all overlaid pixel coordinates use the same transform. The resulting top direction is $-e_\theta / +z/\text{north}$ and the bottom direction is $+e_\theta / -z/\text{south}$.

1.4 Inputs and outputs

Role	Implemented item
Inputs	Black-hole spin, observer distance, inclination, azimuth, FOV, image resolution, torus and funnel visual parameters.
Outputs	<code>CameraPreview/hadros3_camera_preview.png</code> , <code>CameraPreview/hadros3_camera_preview_summary.json</code> .

Approximations	Analytic fallback is schematic. CUDA preview is visual camera rendering; candidate overlays remain pinhole proxies.
Provenance	Records backend used, fallback status, whether CUDA was used, and whether full-Kerr preview mode was requested.

Chapter 2

Black Hole and Kerr Geometry

2.1 Metric functions

HADROS3 uses the Kerr metric in Boyer-Lindquist coordinates and in r_g units. The standard metric functions are

$$\Sigma = r^2 + a^2 \cos^2 \theta, \quad (2.1)$$

$$\Delta = r^2 - 2r + a^2, \quad (2.2)$$

$$A = (r^2 + a^2)^2 - a^2 \Delta \sin^2 \theta. \quad (2.3)$$

The outer horizon radius is

$$r_+ = 1 + \sqrt{1 - a^2}. \quad (2.4)$$

The nonzero covariant metric components implemented for the dynamical calculation are

$$g_{tt} = - \left(1 - \frac{2r}{\Sigma} \right), \quad (2.5)$$

$$g_{t\phi} = - \frac{2ar \sin^2 \theta}{\Sigma}, \quad (2.6)$$

$$g_{rr} = \frac{\Sigma}{\Delta}, \quad (2.7)$$

$$g_{\theta\theta} = \Sigma, \quad (2.8)$$

$$g_{\phi\phi} = \left(r^2 + a^2 + \frac{2a^2 r \sin^2 \theta}{\Sigma} \right) \sin^2 \theta = \frac{A \sin^2 \theta}{\Sigma}. \quad (2.9)$$

The corresponding inverse components used by the Hamiltonian geodesic solver are

$$g^{tt} = - \frac{A}{\Sigma \Delta}, \quad (2.10)$$

$$g^{t\phi} = - \frac{2ar}{\Sigma \Delta}, \quad (2.11)$$

$$g^{rr} = \frac{\Delta}{\Sigma}, \quad (2.12)$$

$$g^{\theta\theta} = \frac{1}{\Sigma}, \quad (2.13)$$

$$g^{\phi\phi} = \frac{\Delta - a^2 \sin^2 \theta}{\Sigma \Delta \sin^2 \theta}. \quad (2.14)$$

2.2 ZAMO quantities

The implemented ZAMO lapse and frame-dragging angular velocity are

$$\alpha = \sqrt{\frac{\Sigma\Delta}{A}}, \quad (2.15)$$

$$\omega = \frac{2ar}{A}. \quad (2.16)$$

For a covariant momentum with components p_t and p_ϕ , the ZAMO energy proxy used in the C++ tetrad utilities is

$$E_{\text{ZAMO}} = -\frac{p_t + \omega p_\phi}{\alpha}. \quad (2.17)$$

The local tetrad initialization maps a normalized local spatial direction (n_r, n_θ, n_ϕ) into a contravariant null vector and then lowers the index with $g_{\mu\nu}$. The implementation checks that the resulting null norm is finite and that the ZAMO energy is positive.

2.3 Null invariant

The null condition is

$$g^{\mu\nu} p_\mu p_\nu = 0. \quad (2.18)$$

Both Python reference utilities and the C++ forward-geodesic backend monitor this invariant. The calculation also monitors the Killing quantities

$$E_\infty = -p_t, \quad (2.19)$$

$$L_z = p_\phi, \quad (2.20)$$

which are conserved because the metric is stationary and axisymmetric.

Chapter 3

Torus / Medium

3.1 Analytic density model

The implemented medium model is an analytic torus density field. It is not a hydrodynamical simulation. The density is evaluated as

$$\rho(r, \theta) = \rho_0 \exp \left[-\frac{1}{2} \left(\frac{r - r_{\text{peak}}}{\sigma_r} \right)^2 \right] \exp \left[-\frac{1}{2} \left(\frac{\theta - \pi/2}{\sigma_\theta} \right)^2 \right], \quad (3.1)$$

with the hard radial support

$$\rho(r, \theta) = 0 \quad \text{if} \quad r < r_{\text{inner}} \quad \text{or} \quad r > r_{\text{outer}}. \quad (3.2)$$

The implemented radial width is

$$\sigma_r = \frac{1}{2}(r_{\text{outer}} - r_{\text{inner}}), \quad (3.3)$$

and the angular width is

$$\sigma_\theta = \max(\theta_{\text{half}}, 10^{-6}). \quad (3.4)$$

The configuration parameter `half_opening_angle_deg` is therefore a Gaussian angular width, not a hard angular boundary. This interpretation is recorded in diagnostics as `half_opening_angle_interpretation = gaussian_width_not_boundary`.

Within the hard radial support, the code can apply a configured `density_floor_g_cm3` by taking the maximum of the analytic value and the floor. Outside the radial shell, the density remains exactly zero.

3.2 MediumRenderer

The shared `MediumRenderer` renders the same analytic density model used by the DIS sampler diagnostics. It records

$$\text{medium_renderer_used} = \text{true}, \quad (3.5)$$

$$\text{density_model_has_hard_radial_cut} = \text{true}, \quad (3.6)$$

$$\text{density_model_theta_profile} = \text{gaussian}, \quad (3.7)$$

$$\text{density_model_theta_is_hard_cut} = \text{false}. \quad (3.8)$$

The renderer draws the hard radial cuts and Gaussian angular density levels. Those angular levels are visual density contours, not physical walls.

3.3 Inputs and outputs

Role	Implemented item
Inputs	<code>rho0_g_cm3</code> , <code>r_inner_rg</code> , <code>r_peak_rg</code> , <code>r_outer_rg</code> , <code>half_opening_angle_deg</code> , optional density floor.
Used by	DIS density evaluation, DIS density diagnostics, forward geometry diagnostics, Observer Bridge diagnostic projections.
Approximation	Static analytic density field; no GRMHD evolution; no self-consistent thermal or composition model.
Output semantics	Positive density only inside the hard radial shell; Gaussian tail in θ .

Chapter 4

UHE Source

4.1 Coordinate cone sampling

The implemented UHE source is a coordinate-volume sampler in a polar cone. For a cone spanning $r_{\min} \leq r \leq r_{\max}$ and $\theta_{\min} \leq \theta \leq \theta_{\max}$, the coordinate volume used for sampling is

$$V_{\text{cone}} = \frac{2\pi}{3} (r_{\max}^3 - r_{\min}^3) (\cos \theta_{\min} - \cos \theta_{\max}). \quad (4.1)$$

Uniform coordinate-volume sampling is implemented by drawing $u_r, u_\theta, u_\phi \sim U(0, 1)$ and setting

$$r = [r_{\min}^3 + u_r(r_{\max}^3 - r_{\min}^3)]^{1/3}, \quad (4.2)$$

$$\cos \theta = \cos \theta_{\min} - u_\theta(\cos \theta_{\min} - \cos \theta_{\max}), \quad (4.3)$$

$$\phi = 2\pi u_\phi. \quad (4.4)$$

The source sampling and physical PDFs are both recorded as

$$p_{\text{source}} = \frac{1}{V_{\text{cone}}}, \quad (4.5)$$

so the implemented source weight is unity for the default model:

$$w_{\text{source}} = \frac{p_{\text{physical}}}{p_{\text{sampling}}} = 1. \quad (4.6)$$

4.2 Energy and directions

The implemented source energy model is monoenergetic:

$$E_{\nu, \text{emit}} = E_0. \quad (4.7)$$

For the isotropic local direction model, the code samples a unit vector in a ZAMO orthonormal frame:

$$\cos \alpha = 2u_\alpha - 1, \quad (4.8)$$

$$\beta = 2\pi u_\beta, \quad (4.9)$$

$$n_r = \cos \alpha, \quad (4.10)$$

$$n_\theta = \sin \alpha \cos \beta, \quad (4.11)$$

$$n_\phi = \sin \alpha \sin \beta. \quad (4.12)$$

The sampling PDF is

$$p_{\Omega} = \frac{1}{4\pi}, \quad (4.13)$$

and the default direction weight is unity.

The source stage does not itself produce the physical Kerr four-momentum for forward propagation. It records the source position, energy, and local direction. The forward-geodesic stage constructs the physical Kerr null momentum from those quantities.

4.3 Inputs and outputs

Role	Implemented item
Inputs	Cone geometry, sample count, random seed, monoenergetic neutrino energy, direction model.
Outputs	<code>UHEsource/uhe_neutrino_source_samples.jsonl</code> and source summary products.
Weights	<code>source_weight = 1</code> and <code>direction_weight = 1</code> for the implemented default uniform/properly matched sampling.
Approximation	Coordinate cone volume, not a GR invariant emissivity integral.

Chapter 5

Forward Geodesics

5.1 Momentum construction

The forward stage consumes UHE source samples and constructs a Kerr null covector. For a local isotropic direction, the ZAMO tetrad maps the local direction into a null four-vector and then into covariant momentum components. For legacy coordinate radial directions, the implementation constructs a covariant momentum with $p_t = -E$ and solves the null constraint for p_r .

The implementation records that the physical Kerr four-momentum is constructed in the forward stage, not in the source sampler.

5.2 Hamiltonian system

The Hamiltonian for null geodesics is

$$H(x, p) = \frac{1}{2} g^{\mu\nu}(x) p_\mu p_\nu. \quad (5.1)$$

The implemented equations of motion are

$$\frac{dx^\mu}{d\lambda} = g^{\mu\nu} p_\nu, \quad (5.2)$$

$$\frac{dp_i}{d\lambda} = -\frac{1}{2} \partial_i g^{\mu\nu} p_\mu p_\nu, \quad i \in \{r, \theta\}. \quad (5.3)$$

The cyclic momenta are conserved by construction in the differential system:

$$\frac{dp_t}{d\lambda} = 0, \quad \frac{dp_\phi}{d\lambda} = 0. \quad (5.4)$$

The production C++ implementation uses an adaptive Runge–Kutta–Fehlberg stepper with invariant checks; the Python reference uses RK4 substeps and may renormalize p_r to keep the null constraint within tolerance. Derivatives of inverse metric components are evaluated according to the configured backend. The solver stops when it reaches the horizon neighborhood, the configured outer radius, maximum steps, or an invalid invariant condition.

5.3 Segment quantities

Each accepted forward path is written as a set of path segments. The adaptive integrator records the actual accepted affine interval $\Delta\lambda_i$ and the momentum normalization E_{norm} used to evolve

the dimensionless covector. For medium four-velocity u^μ , the matter-frame path length is

$$\Delta\ell_{\text{com},i} = \frac{-u^\mu p_\mu}{E_{\text{norm}}} \Delta\lambda_i r_g. \quad (5.5)$$

The contraction and cross-section energy are evaluated at the same segment midpoint. The ratio by E_{norm} accounts for the serialized GeV momentum versus the dimensionless momentum used by the affine integrator. Consequently, $p_\mu \rightarrow cp_\mu$ together with $\Delta\lambda \rightarrow \Delta\lambda/c$ leaves $\Delta\ell_{\text{com}}$ unchanged. The older Boyer–Lindquist spatial-slice distance remains in the segment record only as a geometric diagnostic and is not used for optical depth.

5.4 Inputs and outputs

Role	Implemented item
Inputs	UHEsource/uhe_neutrino_source_samples.jsonl, black-hole spin, step settings, tolerances, outer radius, requested number of samples.
Outputs	ForwardGeodesics/uhe_neutrino_forward_paths.jsonl, ForwardGeodesics/uhe_neutrino_forward_path_segments.jsonl, and summary diagnostics.
Invariants	Null norm, Killing energy, and L_z are monitored.
Approximation	Midpoint evaluation of the covariant matter-frame path length and local energy over each finite affine segment.

Chapter 6

DIS Interaction Sampler

6.1 Density and local energy

The DIS sampler consumes forward path segments and evaluates the analytic torus density at segment midpoints. The baryon density is

$$n_b(r, \theta) = \frac{\rho(r, \theta)}{m_b}, \quad (6.1)$$

where m_b is the baryon mass in grams.

The local neutrino energy is computed from the covariant momentum. If the configured static medium model is valid at that position, the code uses the static observer expression

$$E_{\text{local}} = -p_t u^t, \quad u^t = \frac{1}{\sqrt{-g_{tt}}}. \quad (6.2)$$

If a static observer is not valid, or if the implementation needs the fallback, the ZAMO expression is used:

$$E_{\text{local}} = -\left(\frac{p_t}{\alpha} + \frac{\omega p_\phi}{\alpha}\right) = -\frac{p_t + \omega p_\phi}{\alpha}. \quad (6.3)$$

The static-to-ZAMO fallback is recorded as a physics-risk approximation.

6.2 Cross-section interpolation

The DIS charged-current cross sections are read from compact tabulated files for the implemented GBW and IIM models. If an energy lies inside the tabulated range, the interpolation is log-log:

$$\ln \sigma_{\nu N}(E) = \ln \sigma_0 + \frac{\ln E - \ln E_0}{\ln E_1 - \ln E_0} (\ln \sigma_1 - \ln \sigma_0). \quad (6.4)$$

Requests outside the tabulated range are rejected for that segment and the segment contribution is set to zero in the diagnostic recomputation.

6.3 Optical depth and probability

For each segment,

$$\Delta\tau_i = n_{b,i} \sigma_{\nu N}(E_{\text{local},i}) \Delta\ell_{\text{com},i}, \quad (6.5)$$

where $E_{\text{local},i} = -u^\mu p_\mu$ and $\Delta\ell_{\text{com},i}$ is converted from r_g to cm. This is the discrete midpoint approximation to the invariant integral

$$\tau = \int n_b \sigma_{\nu N}(-u \cdot p) \frac{-u \cdot p}{E_{\text{norm}}} d\lambda r_g. \quad (6.6)$$

The total optical depth along a path is

$$\tau = \sum_i \Delta\tau_i. \quad (6.7)$$

The interaction probability is

$$P_{\text{int}} = 1 - \exp(-\tau), \quad (6.8)$$

implemented numerically as

$$P_{\text{int}} = -\text{expm1}(-\tau) \quad (6.9)$$

with clipping to the interval $[0, 1]$.

6.4 Interaction sampling

If a path is accepted by the Bernoulli trial with probability P_{int} , the segment index is sampled by the inverse CDF

$$P(i) = \frac{\Delta\tau_i}{\sum_j \Delta\tau_j}. \quad (6.10)$$

The Bernoulli trial is evaluated for every path, independent of `max_interactions`. If the number of preliminary successes exceeds that cap, HADROS3 assigns each success an independent deterministic hash priority from the configured seed and event identifier, and retains the smallest priorities. Conditional on the preliminary successes this is a uniform subsample without replacement. It is reproducible and invariant under a permutation of the input path order.

After the segment is selected, let $u \in [0, 1]$ linearly interpolate its coordinates. The midpoint local energy, cross section, and affine conversion are held fixed within that segment, while the medium density is evaluated at 129 nodes. A 128-cell trapezoidal quadrature constructs the intrasegment CDF proportional to

$$\frac{d\tau_i}{du} \propto n_b(r(u), \theta(u)) \sigma_{\nu N}(E_{\text{local},i}) \frac{d\ell_{\text{com},i}}{du}. \quad (6.11)$$

The density is linearly interpolated inside each quadrature cell and its local integral is inverted numerically. Thus the location follows the implemented optical-depth density rather than a uniform coordinate draw. The sampler also requires the final recorded interaction point to have positive medium density and records

$$\text{interaction_point_density_checked} = \text{true}, \quad (6.12)$$

$$\text{interaction_point_inside_medium} = \text{true} \quad \text{when} \quad \rho(r_{\text{int}}, \theta_{\text{int}}) > 0, \quad (6.13)$$

$$\text{interaction_point_rho_g_cm3} = \rho(r_{\text{int}}, \theta_{\text{int}}). \quad (6.14)$$

Rare boundary/fallback behavior is recorded in `interaction_point_sampling_method`. The present criterion for a valid accepted point is $\rho(r_{\text{int}}, \theta_{\text{int}}) > 0$.

6.5 Weights

The DIS products preserve source and direction weights from the UHE source. The pre-event weight used downstream is recorded as `final_pre_event_weight`. The DIS sampler also records optical-depth and expected-interaction quantities so that later stages can distinguish the sampling decision from pre-event observation scoring.

6.6 Inputs and outputs

Role	Implemented item
Inputs	Forward geodesic paths and segments, source samples, DIS cross-section tables, analytic medium parameters, random seed.
Outputs	DIS/dis_path_optical_depths.jsonl, DIS/dis_accepted_interactions.jsonl, optical-depth report, diagnostics, GBW/IIM comparison products.
Diagnostics	Tau distribution, interaction probability distribution, optical depth maps, medium density map, accepted interaction distributions, sigma and correlation plots.
Approximation	Static or ZAMO medium four-velocity; midpoint quadrature of density, local energy, cross section, and covariant comoving path length.

Chapter 7

Observer Bridge

7.1 Purpose

The Observer Bridge consumes accepted DIS interactions and assigns a non-destructive observation score. It does not generate secondary particles, does not call POWHEG, does not call PYTHIA, does not call GEANT4, and does not perform photon transport. All accepted DIS interactions become Observer Bridge candidates, even if they are outside the camera FOV or have small score.

7.2 Geometry proxies

The interaction position is converted to Cartesian coordinates through

$$(x, y, z) = (r \sin \theta \cos \phi, r \sin \theta \sin \phi, r \cos \theta). \quad (7.1)$$

The observer is placed at the configured camera position. The geometric direction from the interaction to the observer is

$$\hat{\mathbf{n}}_{\text{obs}} = \frac{\mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{int}}}{|\mathbf{x}_{\text{obs}} - \mathbf{x}_{\text{int}}|}. \quad (7.2)$$

The implemented escape proxy compares this direction with the local radial direction,

$$w_{\text{esc}} = \max(0, \hat{\mathbf{x}}_{\text{int}} \cdot \hat{\mathbf{n}}_{\text{obs}}). \quad (7.3)$$

This is a proxy, not a particle-transport escape calculation.

The FOV weight is either a hard FOV flag or a soft Gaussian-like angular proxy, depending on configuration. Visibility, line of sight, and redshift are not physical models in H3-W8: all three are immutable unity diagnostic factors, recorded as `unity_diagnostic_not_physics` with their corresponding `*_physics_model=false`. The legacy enable/check flags are retained for configuration compatibility but do not change these unit weights. In particular, no metric redshift integral or occultation test is implied.

7.3 Score definition

The implemented physics weight in the C++ Observer Bridge is

$$w_{\text{physics}} = \text{final_pre_event_weight}, \quad (7.4)$$

with a fallback to source times direction weight when that field is absent. The observer weight is the product of the implemented proxy factors:

$$w_{\text{observer}} = w_{\text{esc}} w_{\text{visibility}} w_{\text{FOV}} w_{\text{distance}} w_{\text{redshift}} w_{\text{LOS}}. \quad (7.5)$$

The final observation score is

$$S_{\text{obs}} = w_{\text{physics}} w_{\text{observer}}. \quad (7.6)$$

All three quantities are recorded for each candidate: `physics_weight`, `observer_weight`, and `final_observation_score`. The implementation enforces non-negative proxy weights by clipping or construction.

7.4 Proxy risk flags

The reports record the proxy nature of the bridge:

$$\text{secondary_particle_proxy_model} \neq \text{full_transport}, \quad (7.7)$$

$$\text{escape_proxy_physics_risk} = \text{true}, \quad (7.8)$$

$$\text{visibility_proxy_physics_risk} = \text{true}, \quad (7.9)$$

$$\text{redshift_proxy_physics_risk} = \text{true}. \quad (7.10)$$

The stage status is scoring-only. It is intentionally non-destructive.

7.5 Inputs and outputs

Role	Implemented item
Inputs	DIS/dis_accepted_interactions.jsonl and runtime configuration. It may consult previous products for diagnostics.
Outputs	ObserverBridge/observer_bridge_candidates.jsonl, ObserverBridge/observer_bridge_ranked_events.jsonl, JSON/CSV summaries, camera-view and overlay diagnostics.
Invariant policy	Number of candidates scored equals number of accepted DIS interactions consumed. FOV is not used as a destructive filter.
Approximation	Geometric escape, visibility, FOV, redshift, and line-of-sight proxies. No secondary-particle generation.

Chapter 8

Observer Image Branch Analysis

H3-W8b introduces an explicit observational image-branch analysis between the Observer Bridge and POWHEG. The Observer Bridge remains a non-destructive candidate scorer: it maps every accepted DIS interaction into an observational candidate and records a closest Kerr-ray diagnostic. H3-W8b asks a different question. Instead of assigning

$$\text{candidate} \rightarrow \arg \min_{\text{ray}} d_{\text{closest}}, \quad (8.1)$$

it groups all compatible camera rays into observed image branches and selects a primary branch by an auditable proxy score. The dashboard presents this stage visually as **Gravitational Image Analysis**, while the legacy `ObserverImageBranches/` directory and base file names remain the compatibility contract.

8.1 Inputs

The implemented stage consumes official Observer Bridge products:

$$\text{ObserverBridge/observer_candidate_kerr_pixel_map.jsonl}, \quad (8.2)$$

$$\text{ObserverBridge/observer_bridge_selected_candidates.jsonl}, \quad (8.3)$$

$$\text{ObserverBridge/observer_bridge_ranked_events.jsonl}. \quad (8.4)$$

When available, it also consumes the Observer Bridge multi-image audit product `ObserverBridge/candidate_m` which records sampled camera rays passing within the configured matching radius of each candidate. No DIS interaction, forward geodesic, camera preview, or Kerr integrator result is modified by H3-W8b.

8.2 Image branch definition

For a candidate c , let $\mathcal{R}_c$ be the set of sampled camera rays whose minimum spatial separation from the candidate is below the matching tolerance. Rays are clustered in image-pixel space. Each connected cluster is an image branch: two rays are adjacent when their Euclidean pixel separation is no greater than the configured link radius, and a branch is a connected component of this undirected graph. Consequently the relation is transitive and the result is invariant under input-ray ordering. Each branch $B_{c,j}$ records:

$$B_{c,j} = \{N_{\text{rays}}, \bar{x}_{\text{pix}}, \bar{y}_{\text{pix}}, \text{Cov}(x_{\text{pix}}, y_{\text{pix}}), \bar{d}_{\text{closest}}, d_{\text{min}}, d_{\text{max}}, s_{\text{pix}}\}. \quad (8.5)$$

Here N_{rays} is the number of compatible rays, $(\bar{x}_{\text{pix}}, \bar{y}_{\text{pix}})$ is the branch centroid on the camera preview pixel plane, and s_{pix} is the RMS pixel spread around the centroid.

8.3 Branch score proxy

The primary observed branch is not selected by the single closest ray. Instead, H3-W8b assigns each branch a proxy score

$$S_{\text{branch}} = w_{\text{rays}} w_{\text{closest}} w_{\text{compactness}}, \quad (8.6)$$

with the implemented proxy weights

$$w_{\text{rays}} = N_{\text{rays}}, \quad (8.7)$$

$$w_{\text{closest}} = \frac{1}{\max(\bar{d}_{\text{closest}}, \epsilon)}, \quad (8.8)$$

$$w_{\text{compactness}} = \frac{1}{\max(s_{\text{pix}}, 1 \text{ pixel})}. \quad (8.9)$$

This score favors image branches supported by many rays, small mean closest approach, and compact clusters on the image plane. It is a ranking proxy, not a full magnification, radiative transfer, or detector-response calculation. Reports and provenance explicitly mark this with `branch_score_is_proxy=true` and `branch_score_not_true_magnification=true`.

8.4 Primary branch selection

For each candidate, H3-W8b selects

$$B_{c,\text{primary}} = \arg \max_j S_{\text{branch}}(B_{c,j}). \quad (8.10)$$

The selected primary branches are written to `ObserverImageBranches/observer_image_primary_branches.jsonl`. Downstream POWHEG preparation consumes that file, so POWHEG receives one primary observed image branch per selected Observer Bridge candidate and does not perform its own image-branch selection.

8.5 Outputs and provenance

The stage writes exclusively under `ObserverImageBranches/`: `observer_image_branches.jsonl`, `observer_image_primary_branches.jsonl`, `observer_image_branch_summary.json`, `observer_image_branch_statistics.json`, diagnostic PNGs, a CSV table, and `observer_branch_view.html`. Provenance records `observer_image_branch_analysis_invoked=true`, `gravitational_image_analysis_invoked=true`, `visual_stage_name=Gravitational Image Analysis`, `branch_scoring_model=ray_count_closeness_compatibility`, `primary_branch_selection_model=argmax_branch_score`, `mean_branches_per_candidate`, `fraction_multi_image_branches`, and `primary_branch_selection_proxy=true`. The stage also writes `gravitational_image_*` explanatory products for multiplicity, score decomposition, primary selection, and POWHEG forwarding without changing the selection algorithm.

When no multi-image audit row exists but a closest-pixel map is available, the compatibility layer may emit one synthetic single-ray branch. Such a record is always tagged `multiplicity_audited=false` and `branch_is_synthetic_fallback=true`; it is not counted as evidence for a single-image multiplicity claim. Conversely, an audit row that explicitly contains zero clusters remains a zero-branch result and is never replaced by the synthetic fallback. These flags are preserved in the primary downstream record and in provenance.

8.6 Physical limitations

The current branch score is observationally motivated but approximate. It does not compute rigorous Kerr caustic magnification, phase-space beam Jacobians, secondary-particle emission, absorption, or detector response. It should be treated as a transparent primary-image proxy until a later stage implements a more complete lensing and transport model.

Chapter 9

POWHEG Hard Process Preparation and Smoke Execution

9.1 Hard subprocess: charged-current neutrino DIS

The implemented POWHEG process is `nudis` from POWHEG-BOX-RES. HADROS3 exposes the perturbative order explicitly. The default `perturbative_order=L0` is a Born-only validation/smoke configuration and writes `L0events` 1. With `perturbative_order=NL0`, that flag is absent and POWHEG evaluates the real and virtual contributions and generates the hardest QCD emission. The Born-level hard subprocess is

$$\nu_\ell(k) + q_i(p) \longrightarrow \ell^-(k') + q_j(p'), \quad (9.1)$$

mediated by a virtual W^+ boson with four-momentum transfer $q^\mu = k^\mu - k'^\mu$. The incoming neutrino carries the full UHE energy $E_{\nu,\text{local}}$ from the accepted DIS interaction point; the target is a parton q_i drawn from the nucleon PDF with momentum fraction x . HADROS3 specifies the two beam particles as

$$\text{ih1} = 12 \quad (\nu_e, \text{PDG electron neutrino}), \quad (9.2)$$

$$\text{ih2} = 1 \quad (\text{proton/nucleon, PDF target}). \quad (9.3)$$

The CC process is selected in the POWHEG input card by `channel_type=3`. The `vtype` input controls photon/Z content for neutral-current channels in this backend and does not select the CC lepton flavor; it is therefore not used by HADROS3 to infer CC flavor.

9.2 DIS kinematic variables

The standard DIS kinematic invariants are built from the neutrino four-momentum k^μ , the nucleon four-momentum P^μ , and the momentum-transfer four-momentum $q^\mu = k^\mu - k'^\mu$:

$$s = (k + P)^2 \approx 2 m_N E_{\nu,\text{local}}, \quad (9.4)$$

$$Q^2 = -q^2 > 0, \quad (9.5)$$

$$x = \frac{Q^2}{2 P \cdot q}, \quad (9.6)$$

$$y = \frac{P \cdot q}{P \cdot k} = \frac{Q^2}{xs}, \quad (9.7)$$

$$W^2 = (P + q)^2 = m_N^2 + Q^2 \frac{1-x}{x}. \quad (9.8)$$

The approximation for s in equation (9.4) holds at UHE energies where $E_{\nu,\text{local}} \gg m_N$. The Bjorken variable $x \in (0, 1]$ is the momentum fraction of the struck parton, $y \in (0, 1]$ is the inelasticity (fractional energy transfer to the hadronic system in the nucleon rest frame), and W is the invariant mass of the produced hadronic system.

9.3 CC neutrino–nucleon DIS cross section

The double-differential cross section for inclusive $\nu_\ell N$ CC DIS is expressed through three structure functions F_2 , F_L , and xF_3 :

$$\frac{d^2\sigma^{\text{CC}}}{dx dy} = \frac{G_F^2 s}{4\pi} \left(\frac{m_W^2}{m_W^2 + Q^2} \right)^2 [Y_+ F_2(x, Q^2) - y^2 F_L(x, Q^2) + Y_- xF_3(x, Q^2)], \quad (9.9)$$

where G_F is the Fermi constant, $m_W \simeq 80.4 \text{ GeV}$ is the W boson mass, and the inelasticity factors are

$$Y_\pm = 1 \pm (1 - y)^2. \quad (9.10)$$

The sign in front of xF_3 is $+Y_-$ for neutrinos (as written) and $-Y_-$ for antineutrinos. The propagator factor $(m_W^2/(m_W^2 + Q^2))^2$ becomes significant when $Q^2 \gtrsim m_W^2$.

At leading order in QCD the structure functions are expressed through the parton distributions weighted by CKM elements. For CC νN DIS the struck parton must be a down-type quark $q_i \in \{d, s, b\}$ (or an up-type antiquark from the sea), and the relevant combinations are

$$F_2^{\text{CC}}(x, Q^2) = 2x \sum_i |V_{q_i q_j}|^2 f_{q_i}(x, Q^2), \quad (9.11)$$

$$xF_3^{\text{CC}}(x, Q^2) = 2x \sum_i |V_{q_i q_j}|^2 \text{sign}(q_i) f_{q_i}(x, Q^2), \quad (9.12)$$

where $V_{q_i q_j}$ is the CKM element for the transition $q_i \rightarrow q_j$ and $\text{sign}(q_i) = +1$ (-1) for quarks (antiquarks). The longitudinal structure function $F_L = F_2 - 2xF_1$ vanishes at LO (Callan–Gross relation) but receives $\mathcal{O}(\alpha_s)$ NLO corrections.

9.4 The POWHEG method

In the configured NLO mode, POWHEG generates the first (hardest) QCD emission with NLO accuracy and provides it as input to a subsequent parton shower. The starting point is the NLO-corrected Born weight

$$\bar{B}(\Phi_n) = B(\Phi_n) + V(\Phi_n) + \int d\Phi_{\text{rad}} C(\Phi_{n+1}), \quad (9.13)$$

where Φ_n is the n -body Born phase space, $B(\Phi_n)$ is the Born squared amplitude, $V(\Phi_n)$ is the renormalized one-loop virtual correction, and $C(\Phi_{n+1})$ is the integrated infrared counterterm that removes double-counting with the real emission. The POWHEG master formula for the fully-differential cross section is

$$d\sigma_{\text{POWHEG}} = \bar{B}(\Phi_n) d\Phi_n \left[\Delta(\Phi_n, p_{T,\text{min}}) + \sum_\alpha \frac{R_\alpha(\Phi_{n+1})}{B(\Phi_n)} \Delta(\Phi_n, k_{T,\alpha}) d\Phi_{\text{rad},\alpha} \right], \quad (9.14)$$

where the sum is over all QCD singular regions α , R_α is the real squared amplitude in region α , and $k_{T,\alpha}$ is the emission transverse momentum in that region. The POWHEG Sudakov form factor is

$$\Delta(\Phi_n, p_T) = \exp \left[- \sum_\alpha \int_{p_T}^\infty \frac{R_\alpha(\Phi_{n+1})}{B(\Phi_n)} d\Phi_{\text{rad},\alpha} \right]. \quad (9.15)$$

The Sudakov factor ensures that the hardest emission is generated with the correct NLO-improved distribution. Events generated by equation (9.14) carry predominantly positive weights—the property that gives POWHEG its name—and reproduce the NLO-accurate inclusive rate after integration.

For the `nudis` process the Born phase space Φ_n covers the two-body $\nu_\ell q_i \rightarrow \ell^- q_j$ kinematics parameterized by (x, y, Q^2) . The NLO real corrections include quark-gluon emission

$$\nu_\ell + q_i \longrightarrow \ell^- + q_j + g \quad (9.16)$$

and the gluon-splitting channel

$$\nu_\ell + g \longrightarrow \ell^- + q_j + \bar{q}_i, \quad (9.17)$$

both of which can produce a gluon or an extra $q\bar{q}$ pair among the outgoing hard-process particles in the LHE record. These terms and channels are disabled in the explicit LO mode.

After the hardest emission is fixed by the POWHEG Sudakov, the parton shower (Pythia 8, not yet invoked in the current HADROS3 pipeline) adds further softer radiation using the POWHEG emission scale as its starting scale.

9.5 Center-of-mass energy and scale

Each accepted DIS interaction vertex in HADROS3 provides the local-frame neutrino energy $E_{\nu,\text{local}}$ to POWHEG as `ebeam1`. The card sets `fixed_target=1`, so `ebeam2` is interpreted as the nucleon mass rather than as the energy of an opposing massless beam. The invariant mass of the neutrino–nucleon hard system is

$$\sqrt{s} = \sqrt{2m_N E_{\nu,\text{local}}}. \quad (9.18)$$

For the UHE range $E_{\nu,\text{local}} \sim 10^9\text{--}10^{12}$ GeV, this gives $\sqrt{s} \sim 43\text{--}1400$ TeV. The upper kinematic bound on the virtuality integration is

$$Q_{\text{max}} = \min\left(\sqrt{2m_N E_{\nu,\text{local}}}, 10^5 \text{ GeV}\right), \quad (9.19)$$

and the lower bound is fixed at $Q_{\text{min}} = 10$ GeV. The cap at 10^5 GeV prevents integration beyond kinematic regions where the fixed-order NLO approximation is reliable. Renormalization and factorization scales are set equal to the virtuality Q at each phase-space point (`runningscales=1`):

$$\mu_R = \mu_F = Q. \quad (9.20)$$

9.6 Parton distributions and CKM flavor structure

HADROS3 uses NNPDF31_nlo_as_0118 (LHAPDF set 303400) for both incoming beams. This is perturbatively matched to the NLO mode; in the LO smoke mode it is retained for controlled pipeline continuity and is explicitly not described as an order-matched LO prediction. The strong coupling α_s is taken from the same PDF set at each renormalization scale (`alphas_from_pdf=1`).

For CC $\nu_\ell N$ DIS the incoming parton must be a down-type quark. The weak transition to an up-type quark is weighted by the squared CKM elements. The dominant channels and their

approximate weights are

$$d \rightarrow u : \quad |V_{ud}|^2 \approx 0.947, \quad (9.21)$$

$$s \rightarrow c : \quad |V_{cs}|^2 \approx 0.947, \quad (9.22)$$

$$b \rightarrow t : \quad |V_{tb}|^2 \approx 1.0 \quad (\text{kinematically suppressed by } m_t \approx 173 \text{ GeV}). \quad (9.23)$$

Off-diagonal CKM contributions ($d \rightarrow c$, $s \rightarrow u$, etc.) are suppressed by the Cabibbo angle and are included by POWHEG but are numerically subdominant. Antiquark sea contributions, including $\bar{u} \rightarrow \bar{d}$ and $\bar{c} \rightarrow \bar{s}$ transitions, also enter through the PDF at small x . At NLO, the gluon PDF contributes through the splitting subprocess $\nu_\ell + g \rightarrow \ell^- + q_j + \bar{q}_i$. The resulting outgoing parton flavors that can appear in the LHE hard-process record are u , c , d , s , and g , with the exact flavor mix determined by PDF support and POWHEG matrix-element channel selection at the given (x, Q^2) .

9.7 LHE hard-process event structure

The Les Houches Event (LHE) file written by `pwhg_main` contains one `<event>` block per generated hard-process realization. Each block records the incoming and outgoing particles with their PDG identifiers, four-momenta (p_x, p_y, p_z, E, m) in GeV, and a status code: -1 for incoming, $+1$ for outgoing. The event header also carries the POWHEG event weight and the hardest-emission scale μ_{POWHEG} that will serve as the starting scale for the subsequent parton shower.

At Born level the event topology is

$$\begin{aligned} \text{status} = -1 : \quad & \nu_\ell + q_i, \\ \text{status} = +1 : \quad & \ell^- + q_j. \end{aligned} \quad (9.24)$$

With the NLO real emission included, the topologies are

$$\begin{aligned} \text{status} = -1 : \quad & \nu_\ell + q_i, & \text{or} & & \text{status} = -1 : \quad & \nu_\ell + g, \\ \text{status} = +1 : \quad & \ell^- + q_j + g, & & & \text{status} = +1 : \quad & \ell^- + q_j + \bar{q}_i. \end{aligned} \quad (9.25)$$

The interpretation of the event weight depends on the LHE IDWTUP strategy and is discussed in the provenance and statistical-weights chapter; the raw sum of `XWGTUP` values is not, in general, the total cross section.

The LHE particles are parton-level hard-process records. In H3-W10 they are the immutable input to PYTHIA 8 showering, hadronization, and decays. GEANT4 transport remains disabled; therefore the H3-W10 output is a generator-level hadronic final state, not a detector-level observable.

9.8 Dry-run role

H3-W9a prepares local POWHEG DIS jobs but deliberately does not execute `pwhg_main`. It consumes primary Observer Image Branch records and writes its own products under `POWHEG/`. It does not modify `ObserverBridge/`, `ObserverImageBranches/`, `DIS/`, `ForwardGeodesics/`, or `UHEsource/`.

The H3-W9a run mode is

$$\text{powheg_run_mode} = \text{dry_run}. \quad (9.26)$$

The reports record

$$\text{powheg_invoked} = \text{false}, \quad (9.27)$$

$$\text{pwhg_main_executed} = \text{false}, \quad (9.28)$$

$$\text{powheg_lhe_generated} = \text{false}. \quad (9.29)$$

H3-W9b adds a controlled real-smoke mode:

$$\text{powheg_run_mode} = \text{real_smoke}. \quad (9.30)$$

This mode executes only the first primary branch record, writes one input card, executes the self-contained local executable `external/powheg/build/DIS/pwhg_main`, and validates a minimal LHE file. It is a pipeline smoke execution, not a precision production campaign. Perturbative order is independent of run mode: both LO and NLO smoke cards are supported and are labeled in summaries and provenance. In H3-W9b itself, PYTHIA, GEANT4, photon transport, and spectra remain disabled; H3-W10 consumes the resulting validated LHE in a separate output directory.

9.9 POWHEG input branch contract

The driver reads `ObserverImageBranches/observer_image_primary_branches.json1`. The selection of downstream candidates is performed by the Observer Bridge and the selection of the primary observed image is performed by H3-W8b. POWHEG does not rank candidates and does not choose among multiple image branches. It prepares one POWHEG request for each primary branch record, except in real-smoke mode where the Python wrapper intentionally executes only the first prepared request as a safety smoke test.

9.10 Input-card mapping

Each selected candidate receives a request identifier of the form `H3PWHG-000001`. Each request has its own `powheg_input_cards/<request_id>/powheg.input`. The implemented mapping is

$$E_{\nu,\text{local}} \rightarrow \text{ebeam1}, \quad (9.31)$$

$$m_N = 0.938272 \text{ GeV} \rightarrow \text{ebeam2}, \quad (9.32)$$

$$\text{events_per_candidate} \rightarrow \text{numevts}, \quad (9.33)$$

$$\text{fixed_target} = 1, \quad (9.34)$$

$$\text{masslesslhe} = 1, \quad (9.35)$$

$$\text{channel_type} = 3, \quad (9.36)$$

$$\text{perturbative_order}=\text{LO} \rightarrow \text{LOevents}=1, \quad (9.37)$$

$$\text{perturbative_order}=\text{NLO} \rightarrow \text{no LOevents flag}. \quad (9.38)$$

The dry-run card also records fixed PDF choices `lhans1 = 303400` and `lhans2 = 303400`. The implemented scale cap is

$$Q_{\text{max}} = \min \left(\sqrt{2(0.938272) E_{\nu,\text{local}}}, 10^5 \right). \quad (9.39)$$

For the implemented seed mode, the seed is

$$\text{powheg_seed} = \text{random_seed} + \text{candidate_rank}. \quad (9.40)$$

9.11 Real-smoke and real-free LHE validation

In H3-W9b real-smoke mode, the prepared POWHEG card is copied into an isolated work directory and `pwheg_main` is executed with that directory as its working directory. The produced `pwgevents.lhe` must satisfy

$$\langle \text{LesHouchesEvents} \in \text{pwgevents.lhe}, \quad (9.41)$$

$$\langle / \text{LesHouchesEvents} \rangle \in \text{pwgevents.lhe}, \quad (9.42)$$

$$N_{\text{LHE events}} > 0. \quad (9.43)$$

The validated LHE and run log are copied to

$$\text{POWHEG/powheg_lhe/H3PWHG-000001/pwgevents.lhe}, \quad (9.44)$$

$$\text{POWHEG/powheg_run_logs/H3PWHG-000001/powheg.log}. \quad (9.45)$$

The validation report records `powheg_invoked=true`, `pwheg_main_executed=true`, `powheg_lhe_generated=true` and `n_lhe_events`. This smoke mode does not modify `ObserverImageBranches/`, `ObserverBridge/`, `DIS/`, `ForwardGeodesics/`, or `UHEsource/`.

H3-W9b real-free mode uses the same local POWHEG execution and LHE validation rules, but it removes the real-smoke safety clamp. The number of interaction candidates is set by the number of records in `ObserverImageBranches/observer_image_primary_branches.jsonl`, and the number of independent POWHEG hard-scattering realizations per selected interaction is set by `events_per_candidate`. Each primary branch record is executed in its own isolated POWHEG work directory. The produced LHE files are copied under `POWHEG/powheg_lhe/<powheg_request_id>/pwgevents.lhe` and the parser aggregates all produced LHE files into the POWHEG particle and event summaries. The provenance records `powheg_real_free_invoked=true`, `real_free_mode=true`, `n_powheg_jobs_requested`, `n_powheg_jobs_run`, `events_per_candidate_requested`, and `n_lhe_events_total`. This mode itself does not invoke PYTHIA, GEANT4, photon transport, or spectra; PYTHIA is invoked only by H3-W10.

9.12 Event requests

The event request file `POWHEG/powheg_event_requests.jsonl` records for each prepared job:

$$\mathcal{R}_{\text{POWHEG}} = \{\text{powheg_request_id}, \text{candidate_rank}, \text{interaction_id}, \text{event_id}, \text{source_sample_id}, E_{\nu, \text{local}}, w_{\text{physics}}, w_{\text{observer}}, S_{\text{obs}}, \text{powheg_input_path}, \text{powheg_seed}, \text{powheg_status}\}. \quad (9.46)$$

The dry-run status is `dry_run_ready`; `powheg_invoked` is false for every dry-run request. In real-smoke mode the initial status is `real_smoke_ready`; after successful validation the request records `real_smoke_lhe_generated`, `powheg_invoked=true`, `pwheg_main_executed=true`, the LHE path, the log path, and `n_lhe_events`. In real-free mode the analogous statuses are `real_free_ready` and `real_free_lhe_generated`; multiple validated LHE products may be recorded in a single POWHEG run.

9.13 Inputs and outputs

Role	Implemented item
------	------------------

Inputs	ObserverImageBranches/observer_image_primary_branches.jsonl and POWHEG configuration values.
Outputs	POWHEG/powheg_event_requests.jsonl, input cards, summary JSON/CSV, report JSON, validation report, di- agnostic plots, and in real-smoke or real-free mode vali- dated pwgevents.lhe products.
Still disabled in H3-W9	No PYTHIA inside the POWHEG stage; no GEANT4; no photon transport; no spectra.
Backend	C++17 executable bin/hadros3_powheg_driver. Python handles orchestration and plots.

Chapter 10

H3-W10 Event Generation with PYTHIA 8 and HepMC3

10.1 Boundary and execution modes

H3-W10 consumes only validated LHE files produced by the H3-W9b `real_smoke` or `real_free` modes. A dry-run card is not an event and is rejected. The implemented transformation is

$$\text{POWHEG LHE} \longrightarrow \text{PYTHIA 8 shower/hadronization/decays} \longrightarrow \text{HepMC3} + \text{JSONL}. \quad (10.1)$$

The `dry_run` mode validates discovery and writes a manifest without calling PYTHIA; `parton_check` parses the LHE and preserves its final partons without a shower; `real_smoke` clamps execution to one request and at most two events; and `real_free` uses the configured positive limits. GEANT4, photon transport, and detector response remain disabled.

Every accepted LHE event maps one-to-one to an event identifier

$$\text{event_generation_event_id} = \text{powheg_request_id} : \text{lhe_event_index}. \quad (10.2)$$

No candidate resampling or implicit oversampling occurs in this stage.

10.2 Fixed-target local frame

The generator frame is the local matter tetrad used to prepare the fixed-target POWHEG card. At ultra-high energy the POWHEG laboratory reboost can leave the incoming lepton energy slightly different from the LHE `EBMUP`. Before external showering, H3-W10 applies one common longitudinal Lorentz boost to every particle of an event,

$$E' = \cosh y E - \sinh y p_z, \quad (10.3)$$

$$p'_z = \cosh y p_z - \sinh y E, \quad (10.4)$$

$$y = \ln(E_{\ell,\text{event}}/E_{\ell,\text{beam}}). \quad (10.5)$$

This maps the incoming massless lepton to the declared beam energy while preserving invariant masses and event topology. After PYTHIA, the complete event is transformed back to the target local energy. Conservation is tested against the physical fixed-target initial state

$$P_{\text{initial}}^\mu = (E_\nu + m_p, 0, 0, E_\nu), \quad m_p = 0.938272 \text{ GeV}, \quad (10.6)$$

with momenta in GeV and HepMC vertices in millimetres.

10.3 Shower, matching, hadronization, and decays

PYTHIA reads the LHE through `Beams:frameType=4`; lepton PDFs are disabled with `PDF:lepton=off`, multiple parton interactions are disabled, and the POWHEG hook vetoes shower emissions above the LHE `SCALUP`. The accepted H3-W10 policy enables final-state radiation, hadronization, and decays, but disables generic initial-state radiation.

This ISR restriction is numerical rather than cosmetic. In the integrated UHE fixed-target smoke, generic PYTHIA ISR produced a relative four-momentum residual of 2.61×10^{-5} , which fails the H3-W10 tolerance. With ISR disabled and POWHEG-vetoed FSR, hadronization, and decays enabled, the measured residual was 9.47×10^{-10} . ISR therefore remains experimental until a DIS fixed-target hook passes the same conservation test. The effective PYTHIA settings, versions, compiler, backend hash, seed, matching policy, and LHE hash are recorded in the manifest and per-job `pythia.cmd` dump.

For each generated event H3-W10 requires

$$\delta_P = \frac{\|P_{\text{final}} - P_{\text{initial}}\|_2}{\max(E_{\text{initial}}, 1 \text{ GeV})} \leq 5 \times 10^{-8}, \quad (10.7)$$

$$\delta_{m^2} \leq 10^{-8}, \quad (10.8)$$

$$Q_{\text{shower,max}} \leq \text{SCALUP} \quad \text{within the backend numerical allowance.} \quad (10.9)$$

Charge conservation is audited, the final hadronized record must contain hadrons, and no final-state quarks or gluons may remain when hadronization is enabled.

10.4 Statistical weights and products

The signed LHE `XWGTUP` value is preserved event by event. Its interpretation follows `IDWTUP`; in particular, H3-W10 does not treat a negative NLO weight as a selection probability and does not assume that the raw weight sum is a cross section. The fields `physics_weight`, `observer_weight`, and `final_observation_score` remain separate metadata. No multiplication of generator and observer weights is performed.

Official products are written only under `EventGeneration/`: immutable input hashes and configuration in `event_generation_manifest.json`, per-job records, aggregate HepMC3, event and final-particle JSONL, validation and particle-content reports, summary JSON/CSV, four diagnostics, and a local event viewer. Rerunning POWHEG invalidates these downstream products, while an H3-W10 run verifies that the upstream LHE hashes did not change.

Chapter 11

Provenance and Statistical Weights

11.1 Stage provenance

Each stage records the backend, products, validation flags, and disabled future stages relevant to its current implementation. A typical provenance record contains:

$$\{\text{parameters, products, validation, stage summaries, disabled stages}\}. \quad (11.1)$$

The implemented provenance reflects the selected execution mode. In a real H3-W10 run it records `pythia_invoked=true` and `expensive_event_generation_invoked=true`; dry-run and parton-check modes leave both false. Later transport stages remain disabled:

$$\text{geant4_invoked} = \text{false}, \quad (11.2)$$

$$\text{photon_transport_invoked} = \text{false}, \quad (11.3)$$

$$\text{spectra_invoked} = \text{false}. \quad (11.4)$$

For H3-W9a, POWHEG is active only as dry-run preparation:

$$\text{status} = \text{powheg_jobs_prepared_no_lhe}. \quad (11.5)$$

For H3-W9b real-smoke execution:

$$\text{status} = \text{powheg_real_smoke_lhe_generated}. \quad (11.6)$$

The provenance also records the scientific theory reference for the run:

$$\text{theory_document} = \text{docs/Theory/HADROS3_Physics_Theory.pdf}, \quad (11.7)$$

$$\text{theory_version} = \backslash\text{TheoryVersion} , \quad (11.8)$$

$$\text{theory_commit} = \text{runtime git commit}, \quad (11.9)$$

$$\text{theory_generation_date} = \text{runtime provenance timestamp}. \quad (11.10)$$

These fields connect each generated result to the theory document that describes the implemented physical and statistical assumptions.

11.2 Weight chain

The implemented source and direction samplers are configured so that the default source and direction weights are unity:

$$w_{\text{source}} = 1, \quad (11.11)$$

$$w_{\text{direction}} = 1. \quad (11.12)$$

The DIS sampler records interaction and pre-event weights. The Observer Bridge then uses

$$w_{\text{physics}} = \text{final_pre_event_weight} \quad (11.13)$$

and computes

$$S_{\text{obs}} = w_{\text{physics}} w_{\text{observer}}. \quad (11.14)$$

The POWHEG dry-run and real-smoke modes carry those weights into each event request. They do not change the physics score. H3-W9a stops after card preparation; H3-W9b executes one local POWHEG smoke job and validates the resulting minimal LHE.

11.3 LHE weight normalization

The parser reads the complete LHE initialization header, including IDWTUP, and each process row

$$(\text{XSECUP}, \text{XERRUP}, \text{XMAXUP}, \text{LPRUP}). \quad (11.15)$$

The generator-declared total cross section is

$$\sigma_{\text{init}} = \sum_{p=1}^{\text{NPRUP}} \text{XSECUP}_p. \quad (11.16)$$

The current POWHEG output uses IDWTUP=-4. For this weighted-event strategy the event-sample estimator recorded by HADROS3 is the signed mean

$$\hat{\sigma}_{\text{event}} = \frac{1}{N} \sum_{i=1}^N \text{XWGTUP}_i, \quad (11.17)$$

with the usual standard error of the sample mean. The raw sum is stored only as a diagnostic and is explicitly marked as not being a cross-section estimator. This distinction makes the estimate invariant if the same event sample is duplicated. For other IDWTUP values, HADROS3 reports σ_{init} and requires the corresponding generator-specific normalization rule rather than silently applying the IDWTUP=-4 formula.

11.4 Random seeds

The implemented random stages record seeds:

- UHE source sampling records its source seed.
- DIS interaction sampling records its acceptance and point-sampling seed.
- POWHEG dry-run and real-smoke modes record `random_seed` and the per-candidate `powheg_seed`.

These seeds are part of the statistical contract of the pipeline.

Chapter 12

H3-W11 GEANT4 Local Material Transport

12.1 Scientific boundary

H3-W11 consumes the H3-W10 HepMC3 final state and transports supported particles through a finite material patch in the orthonormal local matter tetrad. The patch origin is the DIS interaction vertex. Its Cartesian axes are the spatial legs of the local tetrad; energy and momentum use GeV, while HepMC3 and GEANT4 positions use mm. H3-W11 does not integrate Kerr geodesics. The transformation from the local tetrad to the global Kerr frame remains metadata for H3-W12.

Each accepted HepMC3 event maps to one **G4Event**. Final particles map to **G4PrimaryParticle** objects without resampling. Generator, physics, and observer weights remain separate from the initially unit GEANT4 statistical track weight. The official output records local boundary crossings, track and parent identifiers, creator process, position, four-momentum, local time, and boundary normal for later H3-W12 propagation.

12.2 Geometry and material

The implemented geometry is `local_homogeneous_patch_v1`: a box centered on the interaction. Vacuum surrounds the patch and all six boundary faces are audited. In official runs the density is recovered by `interaction_id` from the originating H3-W7 accepted interaction. A configured density is permitted only in explicitly labelled validation fixtures. The baseline composition is a configurable H/He mixture, with mass fractions and density written to the manifest. This is a controlled local-medium prescription, not a laboratory detector and not a full GRMHD toroidal geometry.

The baseline includes no magnetic or electric field and no optical-photon physics. Neutrinos use transport-only pass-through behavior unless an explicitly validated interaction model is later registered.

12.3 Physics list and validity domain

The implemented reference list is `FTFP_BERT` from GEANT4 11.4.2. The H3-W11 v1 contract imposes the conservative upper bound

$$E_{\text{primary}} \leq 10^5 \text{ GeV}. \quad (12.1)$$

This bound follows the documented upper range of the FTF component for the covered hadrons and must not be interpreted as validation above that range. Before `BeamOn`, every final primary is classified by species and energy. An out-of-domain particle produces `unsupported_domain`, writes an audit, and prevents transport. Energy clipping and silent extrapolation are forbidden.

The H3-W10 event audited on 2026-07-13 contained 57 final primaries, of which 22 exceeded the bound. Its largest primary energy was 3.3973576244×10^8 GeV. Consequently the correct H3-W11 result for that event is a domain refusal with zero transported events, rather than an unvalidated GEANT4 shower.

12.4 Scoring and energy ledger

For each event the backend records initial total primary energy, local energy deposit, total energy crossing the patch boundary, steps, and a raw balance. In vacuum the unexplained normalized residual must be below 10^{-6} . Inside matter, rest-mass creation, annihilation, nuclear binding, and the implicit material reservoir prevent the naive identity $E_{\text{in}} = E_{\text{dep}} + E_{\text{escape}}$ from being used as a precision conservation test. H3-W11 therefore records the raw balance separately as an inferred material rest-mass and binding exchange; it is never hidden by renaming the raw quantity.

Numerical validation fixtures test six-axis vacuum crossings, pion decay, photon attenuation in lead, muon stopping-power scaling in water, proton interaction-length scaling in lead, neutrino pass-through, deterministic seeds, signed-weight preservation, atomic publication, and the UHE domain guard. Fixture results validate the implementation inside its declared domain; they do not validate production UHE transport.

12.5 Execution and products

Opening or reloading the GEANT4 web tab never starts transport. Execution requires an explicit POST action or a command-line mode. The official `GEANT4/` directory contains the summary, validation report, environment and physics manifest, import/domain audit, transported event summaries, escaping-particle JSONL, unsupported-particle JSONL, plots, logs, and event viewer. Publication is staged and atomic. A new H3-W10 product invalidates H3-W11 and all later transport products.

The precise implementation status is therefore

H3-W11 implemented and numerically tested for the declared supported-energy domain; real HADROS3 UHE production remains blocked by the physics-domain guard.

Chapter 13

Pipeline Summary

13.1 Implemented stage contract

The current implemented pipeline can be summarized as

UHESource $\rightarrow$ ForwardGeodesics $\rightarrow$ DIS $\rightarrow$ ObserverBridge $\rightarrow$ POWHEG RealSmoke $\rightarrow$ EventGeneration $\rightarrow$ (13.1)

The camera preview is an observer-view diagnostic that can be used by the dashboard and by Observer Bridge overlays, but it is not a physical secondary-particle propagation stage.

Table 13.1: Implemented pipeline products and semantic status.

Stage	Official inputs	Official outputs	Status
UHE Source	Configuration	UHEsource/ JSONL/JSON	Sampling proxy with monoenergetic source.
Forward Geodesics	UHEsource/	ForwardGeodesics/ paths and segments	Kerr null geodesic propagation.
DIS Sampler	ForwardGeodesics/, UHEsource/	DIS/ optical-depth and accepted interactions	Optical-depth DIS sampler.
Observer Bridge	DIS/dis_accepted_interactions	ObserverBridge/ candidates and ranked events	Non-destructive geo- metric scoring proxy.
Gravitational Im- age Analysis	ObserverBridge/ se- lected candidates and Kerr pixel-map prod- ucts	ObserverImageBranches/ branches and primary branches	Observed image branch proxy selec- tion.
POWHEG Real Smoke	ObserverImageBranches/ POWHEG/realSmoke	POWHEG/realSmoke, primary H3-W9b local one- report, and one smoke LHE	H3-W9b local one- job LHE smoke.
Event Generation	Validated real H3-W9b LHE files	EventGeneration/ HepMC3, JSONL, reports, plots, and manifest	H3-W10 PYTHIA 8 FSR, hadronization, decays, and HepMC3 conversion; ISR dis- abled by numerical policy.

Stage	Official inputs	Official outputs	Status
GEANT4	Validated H3-W10 HepMC3 and H3-W7 local density metadata	GEANT4/ audits, mani- fests, boundary parti- cles, ledgers, plots, and logs	H3-W11 local ma- terial transport inside the declared supported-energy domain; UHE inputs are blocked before transport.

13.2 Explicit non-goals of the implemented stages

The current pipeline does not yet implement:

- an accepted fixed-target DIS initial-state shower; generic PYTHIA ISR is experimental and disabled by default after failing the conservation tolerance;
- GEANT4 transport of the current UHE event beyond the declared supported-energy envelope;
- photon transport from secondaries to the observer;
- full GRMHD torus evolution;
- full physical visibility, escape, redshift, or line-of-sight transport.

Those future stages must preserve the contract that each stage consumes official previous outputs, writes only to its own directory, records provenance, and marks expensive or proxy physics explicitly.

Appendix A

Appendix: Symbols and Product Names

Product or field	Meaning
<code>uhe_neutrino_source_samples.json</code>	UHE neutrino source positions, energies, directions, and sampling weights.
<code>uhe_neutrino_forward_path_segments.json</code>	Forward path segments with midpoint coordinates, momenta, and segment lengths.
<code>dis_path_optical_depths.json</code>	Path-level DIS optical depths and interaction probabilities.
<code>dis_accepted_interactions.json</code>	Accepted DIS interactions and their density-checked interaction points.
<code>observer_bridge_candidates.json</code>	Accepted DIS interactions scored as observation candidates.
<code>observer_bridge_ranked_events.json</code>	Accepted observation candidate family ranked by final observation score.
<code>observer_image_primary_branches.json</code>	Primary observed image branch per selected Observer Bridge candidate.
<code>powheg_event_requests.json</code>	Dry-run POWHEG event requests and card paths.
<code>final_pre_event_weight</code>	Physics weight consumed by Observer Bridge as implemented.
<code>final_observation_score</code>	Product of implemented physics and observer weights.
<code>powheg_invoked</code>	False in H3-W9a dry run; true in H3-W9b real smoke.
<code>pwheg_main_executed</code>	False in H3-W9a dry run; true in H3-W9b real smoke.
<code>powheg_lhe_generated</code>	False in H3-W9a dry run; true in H3-W9b real smoke.
<code>event_generation_events.hepmc</code>	Aggregate H3-W10 generator-level event record in HepMC3 ASCII format.
<code>xwgtup</code>	Signed LHE generator weight, kept separate from observer scores.
<code>pythia_invoked</code>	True only in H3-W10 <code>real_smoke</code> or <code>real_free</code> ; false in dry-run and parton-check modes.
<code>geant4_escaped_particles.json</code>	H3-W11 local boundary crossings ready for H3-W12, including lineage, process, local phase space, time, and weight.
<code>geant4_invoked</code>	True only when <code>BeamOn</code> transported an input fully inside the declared domain; false for import checks and UHE refusals.